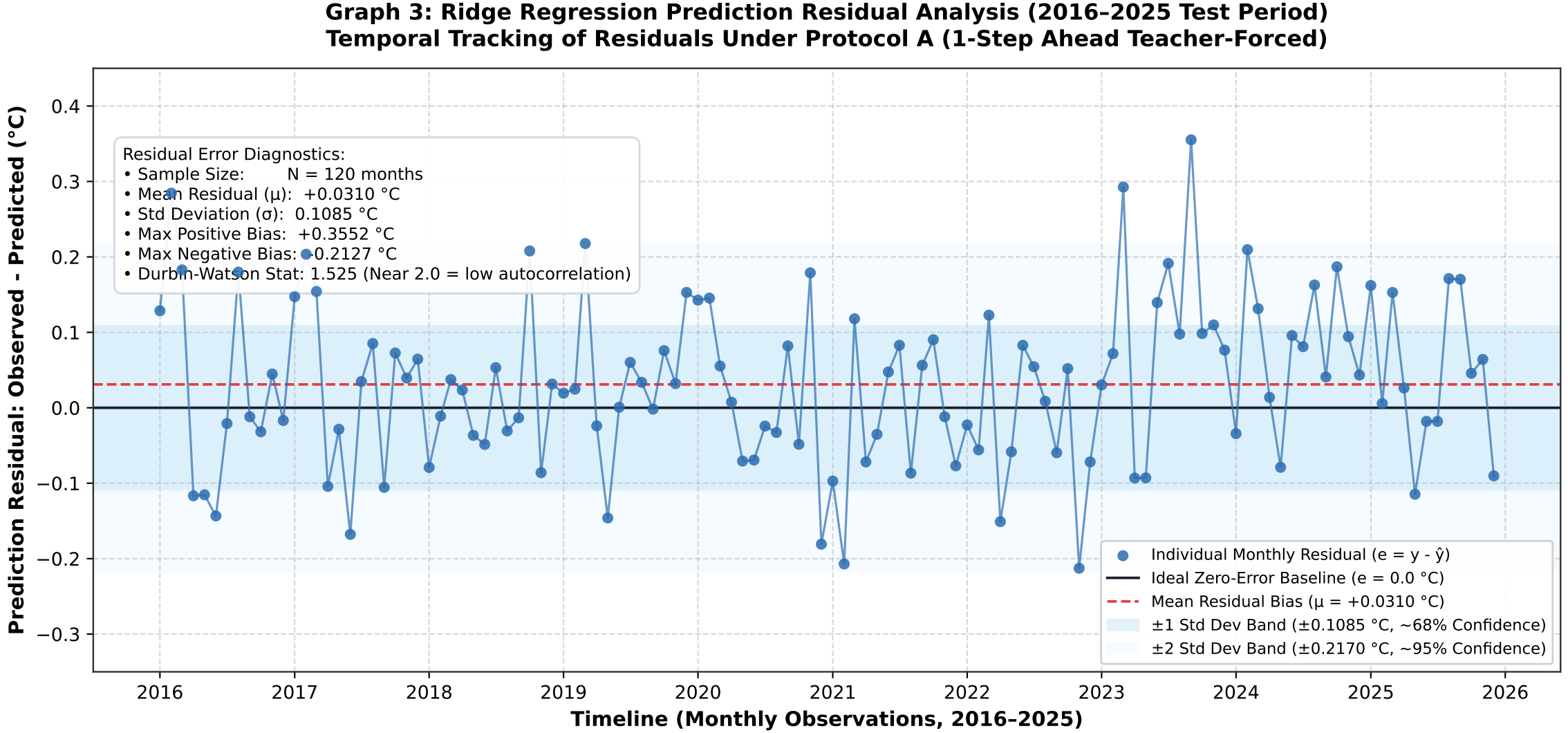
Graph 3: Ridge Regression Prediction Residual Analysis (2016–2025 Test Period)
Temporal Tracking of Residuals Under Protocol A (1-Step Ahead Teacher-Forced)
Residual Error Diagnostics:
• Sample Size: N = 120 months
• Mean Residual (μ): +0.0310 °C
• Std Deviation (σ): 0.1085 °C
• Max Positive Bias: +0.3552 °C
• Max Negative Bias: -0.2127 °C
• Durbin-Watson Stat: 1.525 (Near 2.0 = low autocorrelation)
Individual Monthly Residual (e = y - ŷ)
Ideal Zero-Error Baseline (e = 0.0 °C)
Mean Residual Bias (μ = +0.0310 °C)
±1 Std Dev Band (±0.1085 °C, ~68% Confidence)
±2 Std Dev Band (±0.2170 °C, ~95% Confidence)
Prediction Residual: Observed - Predicted (°C)
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Timeline (Monthly Observations, 2016–2025)